3.1.3 How the body fights disease

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 5			
Infectious (communicable) diseases are caused by microorganisms called pathogens.	Types of pathogen limited to bacteria and viruses.	4.3.1.1	4.3.3.1
These may reproduce rapidly inside the body and may produce poisons (toxins) that make us feel ill. Viruses damage cells in which they reproduce.	No recall of specific illnesses is required.	4.3.1.1	4.3.3.1
Outcome 6			
White blood cells help to defend against bacteria by ingesting them.		4.3.1.6	4.3.3.4
Vaccination involves introducing small quantities of dead or inactive forms of a pathogen into the body to stimulate the white blood cells to produce antibodies so that if the same pathogen re-enters the body, antibodies can be produced rapidly. Students should be able to explain the use of vaccination in the prevention of disease.		4.3.1.7	4.3.3.5
Outcome 7			
Medical drugs are developed and tested before being used to relieve illness or disease. Drugs change the chemical processes in people's bodies. People may become dependent or addicted to the drugs and suffer withdrawal symptoms without them.		4.3.1.9	4.3.3.7

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Antibiotics, including penicillin, are medicines that help to cure bacterial disease by killing infective bacteria inside the body, but cannot be used to kill viruses.	The names of any antibiotics other than penicillin are not required. **Suggested activity for TDA** Use pre-inoculated agar in Petri dishes to evaluate the effect of disinfectants and antibiotics.	4.3.1.8	4.3.3.6